

Veltrixa – Project Case Description

1. Introduction

Artificial intelligence is rapidly transforming the way organizations manage business information. By automating document classification, metadata extraction, and information retrieval, AI-powered document management platforms can significantly improve operational efficiency. At the same time, these technologies introduce important challenges relating to privacy, information security, governance, and regulatory compliance.

Veltrixa is a fictional Business-to-Business (B2B) Software-as-a-Service (SaaS) provider created as a realistic Privacy & Governance, Risk and Compliance (GRC) case study. The purpose of the project is to demonstrate how GDPR and ISO/IEC 27001 requirements can be translated into practical documentation during the design and implementation of a modern cloud-based service.

Although the organization is fictional, the business scenario, system architecture, processing activities, and compliance documentation have been designed to reflect realistic industry practices.

2. Business Scenario

Veltrixa develops an AI-powered document management platform for medium and large organizations that process significant volumes of business documentation.

Typical customers include organizations operating within sectors such as manufacturing, professional services, healthcare, finance, and public administration, where documents often contain confidential or personal information.

The platform enables organizations to:

- securely upload business documents;
- automatically classify documents using artificial intelligence;
- extract searchable metadata;
- perform intelligent document searches;
- manage user permissions;
- maintain comprehensive audit logs.

Because uploaded documents frequently contain personal data, the platform has been designed with privacy, security, and compliance as core architectural principles rather than post-implementation additions.

3. Project Objective

The objective of this project is to establish the core privacy and information security documentation required before deploying a cloud-based SaaS platform that processes personal data.

Rather than focusing solely on legal compliance, the project demonstrates how privacy and security requirements can be integrated throughout the system development lifecycle.

The documentation illustrates a structured approach to:

- understanding personal data processing;
 - identifying privacy risks during solution design;
 - assessing regulatory obligations;
 - documenting processing activities;
 - establishing governance and security controls.
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4. Solution Overview

The fictional solution consists of the following primary components:

Business Users

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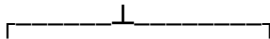
Web Application

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Application API

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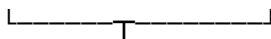


AI Classification Engine

Metadata Database

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Secure Document Storage

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Administrative Portal

Documents uploaded through the web application are transmitted via encrypted connections to the application backend.

The AI classification engine analyses document content and extracts metadata used for search and categorisation. Original documents are securely stored, while metadata is indexed separately to improve search performance.

Administrative users manage platform configuration, user administration, and monitoring through a dedicated administration portal.

5. High-Level Technology Stack

The fictional platform is assumed to use a modern Microsoft Azure-based architecture.

Component	Technology
Frontend	React
Backend API	ASP.NET Core
Identity Provider	Microsoft Entra ID
AI Service	Azure AI Document Intelligence
Database	Azure SQL Database
Document Storage	Azure Blob Storage
Secrets Management	Azure Key Vault
Logging & Monitoring	Azure Monitor / Log Analytics

These technologies have been selected to create a realistic enterprise environment suitable for demonstrating GDPR and ISO/IEC 27001 implementation.

6. Personal Data Processing

Depending on customer usage, uploaded documents may contain:

- Names
- Email addresses
- Telephone numbers
- Postal addresses
- Employment information
- Customer information
- Financial information
- Contractual documentation
- Other personal data contained within business documents

Veltrixa acts as a **data processor**, processing personal data solely on behalf of its customers, who act as **data controllers**.

7. Key Design Assumptions

The following assumptions apply throughout the project.

Topic	Assumption
Service Model	Multi-tenant SaaS
Customer Location	European Union
Data Controller	Customer organisation
Data Processor	Veltrixa
Authentication	Microsoft Entra ID (Enterprise SSO)
Authorisation	Role-Based Access Control (RBAC)
Encryption in Transit	TLS 1.3
Encryption at Rest	AES-256
Logging	Security and audit logging enabled
Data Residency	EU cloud region
AI Usage	Document classification and metadata extraction only

8. Compliance Documentation

The portfolio documents follow the same logical progression as a typical privacy and information security project.

Business Requirements

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System Design

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Data Flow Mapping

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Privacy by Design Review

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DPIA Screening

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Record of Processing Activities (ROPA)

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Information Security Management System (ISMS)

Each document builds upon the previous one, demonstrating how privacy and security considerations evolve throughout the implementation of a new information system.

9. Project Deliverables

Document	Purpose
Data Flow Mapping	Identifies how personal data flows throughout the platform.
Privacy by Design Review	Assesses GDPR Article 25 implementation during system design.
DPIA Screening	Determines whether a full Data Protection Impact Assessment is required.
Record of Processing Activities (ROPA)	Documents processing activities in accordance with Article 30 GDPR.
Information Security Management System (ISMS)	Demonstrates governance and information security controls aligned with ISO/IEC 27001.

10. Applicable Standards

The project has been developed with reference to:

- Regulation (EU) 2016/679 (General Data Protection Regulation)
- ISO/IEC 27001:2022
- ISO/IEC 27002:2022
- Privacy by Design principles
- Risk-based information security management

11. Disclaimer

Veltrixa is a fictional organization created exclusively for educational and portfolio purposes.

The project has been developed to demonstrate practical competencies within Privacy, GDPR, Governance, Risk & Compliance (GRC), and Information Security Management using realistic business scenarios and industry-standard documentation practices.